How Correct Is Your Answer?

A Semantic Correctness Framework for Open QA Evaluation

Elitsa Yotkova, Violeta Kastreva*, Petar Velkov, Hristo Boyanov, Dimitar Dimitrov, Preslav Nakov, Ivan Koychev*

Sofia University • ETH Zurich • University of Zurich • MBZUAI **equal contribution*



Semantic correctness as an ordered taxonomy, with monotonicity as the evaluation criterion.

Why Is Open QA Evaluation Hard?

- Free-form answers often have no explicit correctness label.
- A gold answer gives us a **semantic anchor — not a string to match.**
- Binary **accept**/**reject** judgments hide important distinctions.
- Answers can be partial, overinclusive, unsupported, contradictory, or based on a false premise.

Correctness Is Not Binary

Question	Name 3 cities in Europe
Gold	Paris, Rome, Sofia
Overinclusive invalid	Paris, Berlin, Tokyo
Partial	Oslo, Amsterdam
Overinclusive valid	Paris, Berlin, Sofia all of which have **population over 1.2 million people**



Not every answer is simply correct or wrong—correctness comes in different semantic forms and degrees.

How Should We Classify Correctness Instead?

Correct

1. **Exact** — Prediction is identical to the gold answer.
2. **Equivalent** — Rephrased or semantically equivalent to the gold answer.
3. **Alternative Correct** — Different answer, but still correct under the question.
4. **Overinclusive Valid** — Contains the correct answer plus extra valid information.
5. **Partial** — Contains only part of the required information.
6. **Overinclusive Invalid** — Contains the correct answer plus extra invalid information
7. **Invalid** — Semantically incorrect answer.
8. **Contradictory** — Explicitly contradicts the gold answer or question premise.

Wrong

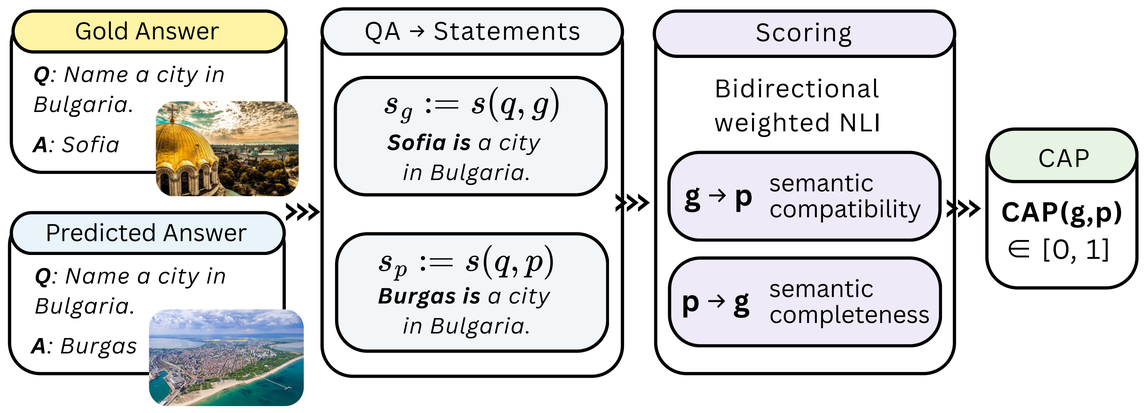
- The gold/reference answer defines the expected content.
- The taxonomy captures the **(1)**relationship between gold and predicted answers and **(2)***common LLM-response patterns*.
- A good metric should **score more correct answers higher than less correct ones**.

Monotonicity Criteria For Correctness

exact* ≈ *equivalent* ≈ *alternative-correct
≥ *overinclusive_valid* > *partial* >
overinclusive_invalid* > *invalid* ≥ *contradictory
More correct classes should receive a higher metric scores

CAP & Our Evaluation Framework

CAP (Context-Aware Precision) is a reference-based metric for evaluating semantic correctness in Open QA.



$$D(s_g \rightarrow s_p) := P_{entailment}(s_g \rightarrow s_p) + \lambda P_{neutral}(s_g \rightarrow s_p)$$

where λ ∈ [0, 1] controls the contribution of the neutral class

$$CAP(s_g, s_p) := \alpha D(s_g \rightarrow s_p) + (1 - \alpha) D(s_p \rightarrow s_g)$$

In our experiments, we use α = 0.85 and λ = 0.30

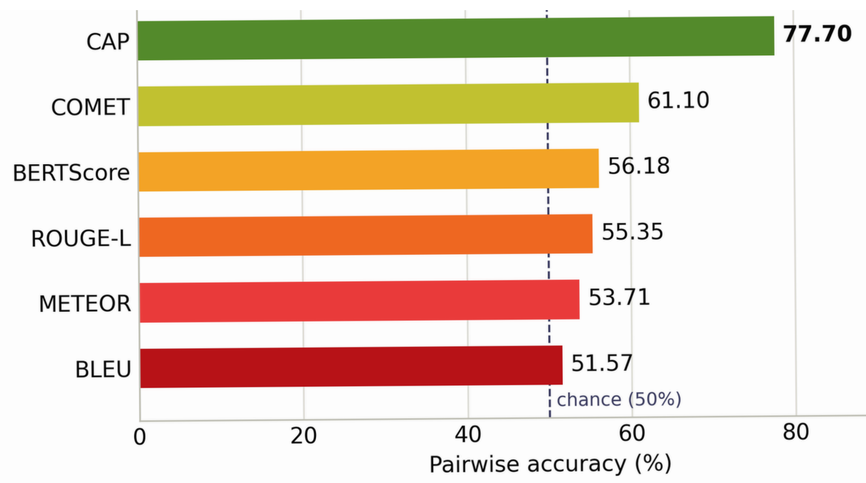
Our Contributions

- **Taxonomy + dataset**: 8 correctness classes, 8.8k examples
- **Metric + reformulation data**: CAP + 11k QA→statement triples
- **Diagnostic protocol**: checks monotonicity criteria for correctness and reveals metric **failure** modes

How do we know whether a metric score reflects correctness?

Gold statement	Prediction statement	relationship	CAP
Water boils at 100°C	Water boils at 0 °C	*contradictory*	CAP 0.01
Water boils at 100°C	The boiling point of water is 100°C	*equivalent*	CAP 0.98

We use our taxonomy as human-interpretable target ordering.



Monotonicity violations

CAP 4/25

COMET 9/25

BertScore 11/25

Evaluators in our study often **fail** to preserve the intended semantic correctness ordering.

Does the taxonomy generalize to real LLM outputs?

In 3,000 responses from 3 LLM families, (1) all eight taxonomy classes appeared; (2) CAP largely preserved semantic ordering

Reusable Resources

CAP-Correctness

8,827

question, gold, predicted, label
8 semantic correctness classes
OpenBookQa ARC MMLU

1,638 human labeled

$\kappa = .779$ human-LLM agreement

CAP-Statements

11,000

question, answer, statement
4 question types
train/dev/test reformulation

1,353 manually checked

90% correct

Paper + Data & Code

Paper

Code & Data